

MARK LORENZ YAMANAKA

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SUMMARY

PRC-licensed Professional Aeronautical Engineer (2024) with six completed aerospace projects spanning CFD, FEA, structural sizing, systems integration and GD&T drawing packages, plus maintenance and ground-support internships across A320, ATR 72, B737 and Cessna 172. Analysis practice is validation-first: hand calculations set the baseline, mesh convergence is proven, and outcomes are published in full, including the two that fell short of target.

SKILLS

Stress & Structural Analysis: Hand calculations, stress linearization, membrane-plus-bending classification, V-n envelope, limit and ultimate load, mesh convergence, sensitivity studies

FEA & CFD: ANSYS Mechanical, ANSYS Fluent, one-way FSI, CalculiX, PrePoMax, OpenFOAM, ParaView, XFOIL

CAD & Drawings: CATIA V6, SolidWorks, ANSYS SpaceClaim, GD&T, tolerance stack-up, BOM, machining route, inspection plans

Standards: FAR Part 23 / 27 / 29, ASME Y14.5-2018, ASME BPVC VIII-1 Appendix 2, PCAR Part 6, FAR Part 145

Documentation: Stress reports, substantiation packages, task cards, work orders, non-routine cards, certificate of release to service

Programming & Tools: Python, MATLAB/Simulink, C++, Git/GitHub, Microsoft Office, Power BI, Asana, OriginLab

Languages: English (Native), Cebuano (Native), Filipino (Professional), Japanese (Conversational)

Certifications: PRC Professional Aeronautical Engineer License, Philippines, December 2024

EXPERIENCE

Aircraft Maintenance Engineer Intern | Dornier Technology Inc.

May 2023 – Jul 2023

- Supported 10+ base and line maintenance checks at a third-party MRO on customer fleets (Cebu Pacific, Royal Air) covering A320, ATR 72-500, B737-200 and BAE 146, under PCAR Part 6 AMO approval, the Philippine equivalent of FAR Part 145.
- Performed 15+ pre- and post-check inspections during line maintenance, then supported an A320 engine overhaul in base maintenance.
- Installed torque-controlled fasteners on an A320 wing leading-edge de-icing boot under a licensed mechanic's sign-off; assisted access panel removal and refit throughout.
- Inspected in-service airframe structure for damage and corrosion; observed and assisted dye penetrant and borescope inspection.
- Handled the records chain for each task: task cards, work orders, non-routine cards raising unplanned findings, and certificate of release to service; documentation closed internal audit with zero non-conformities.
- Reported to licensed aircraft maintenance engineers and base maintenance supervisors.

Ground Support Group Intern | Alpha Aviation Group Philippines

Feb 2023

- Completed flight simulation sessions on a Cessna 172 and an Airbus A320 full-flight simulator during a two-week rotation.
- Performed external inspection and cockpit preparation on trainer aircraft.
- Conducted a supervised engine run-up and taxi on the Cessna 172, working through the function checks and fault indications that accompany them.

Part-Time Leader / Sub-Leader, Packaging and Logistics | ZOZOBASE

Feb 2018 – May 2026

- Led shift operations for a 15-member packaging and logistics team in a Japanese-language environment, briefing floor operators and reporting to supervisors.
- Proposed workflow changes adopted by shift supervisors, raising throughput 20% and cutting cycle time 12%.

PROJECTS

Uncooled SS316 Convergent-Divergent Nozzle | github.com/dmmarkwork-cloud/ss316-cd-nozzle-feasibility

- Carried an uncooled SS316 C-D nozzle ($P_c = 2$ MPa, $T_c = 800$ K) from CFD through thermal-structural analysis into a full GD&T drawing set. One-way FSI mapped Fluent wall temperature and pressure onto the FEA mesh; linearized membrane-plus-bending stress placed the governing location at the chamber-to-converging junction (29.6 MPa) rather than the throat (17.2 MPa), giving FoS 7.12 across a three-level mesh convergence study.
- Passed CFD validation gates at peak wall $y^+ 1.74$, mass imbalance $3.0e-5\%$, discharge coefficient 0.977, and exit Mach 2.85 against 2.94 by hand calculation. Identified creep at a 527 °C wall, not static yield, as the real design driver.

- Sized an 8× M8 A286 bolted joint per ASME BPVC VIII-1 Appendix 2 and detailed the package in CATIA V6 with tolerance stacks closing at 1.68×, machining route, inspection plan and BOM. Published the v1.0 error (FoS 42.70, traced to a wrong assessment location and stress misclassification) alongside the corrected v2.0.

AI 6061-T6 Propellant Tank: Static FEA Verification | github.com/dmmarkwork-cloud/gas-vessel-fea

- Ran static linear FEA of a 3 mm-wall tank with hemispherical caps at 3.25 MPa (inner radius 147 mm, R/t = 49) in CalculiX via PrePoMax using C3D10 elements. Mid-cylinder von Mises 136.6 MPa against 137.91 MPa by thin-wall hand calculation, agreeing to 1.0%, with mesh converged at 0.62% on 264,007 nodes.
- Reported FoS 1.91 against a self-defined 2.0 target at the cylinder-to-hemisphere junction, a 3D geometric effect thin-wall theory does not capture, and recommended a 3.2 mm wall. Documented an April 2026 boundary-condition correction that moved peak stress from 204.1 MPa to 144.5 MPa by relocating constraints off the critical junction.

MK-10 Light Aircraft: FAR Part 23 Design & Structural Sizing | github.com/dmmarkwork-cloud/mk10-aircraft-design

- Held one design baseline from mission definition through weight and CG, drag buildup and the V-n envelope into spar sizing for a two-seat light aircraft (1,649 lb gross weight, 126.5 ft² wing, aspect ratio 6.58, NACA 2415, 125 BHP). Validated 459 nmi range, 7.63 h endurance and 9,536 ft service ceiling.
- Sized front and rear wing spars from a 10-strip spanwise load distribution per Bruhn to a 4.4G utility limit and 7.4G ultimate, carrying every weight change through to the shear-moment diagrams, then resolved the design into control surfaces, landing-gear load cases and a SolidWorks three-view.

PANO-VISTA: Fixed-Wing UAV Design & Systems Integration | github.com/dmmarkwork-cloud/pano-vista-uav

- Modelled a tractor-configuration fixed-wing UAV part by part in one SolidWorks master assembly, then built it with brushless propulsion (A2212/6T 2200KV, 40A ESC, LiPo), an eight-channel receiver and four micro servos across three control axes. Delivered exploded isometric and full orthographic drawings with BOM, integrating vendor CAD alongside custom parts. Five-person team project.

Wooden Rotor Blade: Drawing Package | github.com/dmmarkwork-cloud/rotor-blade-helicopter-design

- Drafted a Ø 71.10 in two-blade helicopter rotor in orthographic projection on tracing paper, then rebuilt it parametrically in SolidWorks as a complete blade and hub assembly with dimensioned three-view and isometric. Nonlinear twist falls from 78° inboard to 24° at the tip so angle of attack holds roughly constant as local velocity climbs with radius; solidity 0.095.

High-Lift Device Demonstrator: Servo-Actuated Wing Section

- Built a sectioned wing in cardboard skinned in aluminium foil on a presentation stand, with servos bedded between the ribs driving the ailerons and leading-edge slats from an Arduino Uno sketch written in C++, operated from a dedicated control interface on the base so the high-lift devices deploy on demand at eye level.

ORGANIZATION

Chairperson / Officer-in-Charge | **SEA-HAU Debate Committee**

Sep 2022 – Nov 2022

- Led a university-wide debate program with 25+ participants, coordinating judges and partner organizations from planning through execution.
- Developed standardized evaluation criteria and topic frameworks to keep judging fair and consistent across the competition.

Point-Person | **SEA-HAU Debate Committee**

Jul 2022 – Aug 2022

- Acted as primary liaison between debate participants and the organizing committee, relaying feedback and enforcing committee policy.
- Guided participants on parliamentary debate procedure and coordinated between teams, adjudicators, and committee members during events.

EDUCATION

Bachelor of Science in Aeronautical Engineering

Jul 2020 – May 2024

Holy Angel University, Angeles City, Philippines

- Honors: Dean's List (GPA 3.6/4.0)
- Relevant Coursework: Aircraft Structures, Aircraft Systems, Aircraft / Helicopter / UAV Design, Aerodynamics, Powerplant, Reliability Engineering, Engineering Data Analysis